# Optimal Parameters for Finite-Key Decoy-State BB84 under the Lim et al. (2014) Bounds: a Verified Optimizer, a Corrected Model, and a Benchmark Protocol for Surrogates

Final report of the `optimizer-jax-rework` branch (supersedes all earlier reports)

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**Abstract**

We compute the parameters that maximise the secret key rate $R$ of finite-key decoy-state BB84 under the security bounds of Lim, Curty, Walenta, Xu and Zbinden (PRA 89, 022307, 2014): signal and decoy intensities, their probabilities and the basis probability, as functions of fiber length $L \in [0, 200]$ km and block size $n_X \in [10^4, 10^9]$.

An audit of the earlier pipeline found four problems:

- the earlier key-rate code used the $X$-basis detection count where Eq. (4) of the paper requires the $Z$-basis error count;
- it departed from the paper in four further model settings;
- the optimizers that produced the neural-network training labels did not reach the optimum;
- most headline numbers of the earlier reports could not be reproduced.

We present a corrected model and a JAX optimizer whose output is verified against an independently built reference:

- The corrected model agrees with an independent re-implementation to $\leq 2 \times 10^{-13}$ in $\ell$ and reproduces the paper's three stated numbers to within 8 %.
- The optimizer uses an exact Hessian, penalty-free constraints and continuation in $L$. Its relative regret is at most $1.03 \times 10^{-13}$ on the bulk grid and $9.50 \times 10^{-13}$ on a dense grid near the maximal distance, at 100 % of scored points.
- For comparison, the earlier dual-annealing run is within $10^{-6}$ at 3.7 % of points, and the run that produced the neural-network labels at 81.7 % (bulk) and 0.0 % (tail).
- The optimal parameters vary smoothly with $L$, apart from switches between competing local optima, where the parameters jump but $R$ stays continuous. There are at least ten such switches below the maximal distance across the six block sizes (Section 5). No single sweep logged all of them.
- A warm-started single-point solve takes 1199 $\mu$s on a laptop CPU.

All earlier neural-network results are withdrawn. We specify, in advance, the protocol by which a retrained classical network and a quantum machine-learning model will be benchmarked against this reference.

## Contents

**1 Introduction and scope** **2**

**2 Model: finite-key decoy-state BB84 (Lim et al. 2014) 2**
2.1 Channel, detection and error model . . . 2
2.2 Key length (Eqs. (1)–(5) of the paper) . . . 3
2.3 Constants and presets . . . 3
2.4 Choices the paper leaves open . . . 3
2.5 Agreement with the paper . . . 4
2.6 Effect of each correction . . . 4
2.7 Numerical correctness . . . 5

**3 Optimizer 5**

**4 Evaluation protocol 6**
4.1 Trusted reference . . . 6
4.2 Metrics . . . 7

**5 Results 7**
5.1 Horizons and key rate . . . 7
5.2 Accuracy of the optimal parameters and key rates . . . 8
5.3 Structure of the optimal parameters . . . 9
5.4 Computational cost . . . 12

**6 Discussion 13**
6.1 What changed relative to the earlier work . . . 13
6.2 Threats to validity . . . 13
6.3 Open issues . . . 14

**7 Benchmark protocol for surrogates (classical NN and QML) 14**

**8 Conclusion 16**

**A Earlier documents and their status 17**

**B Additional figures 18**

**C Reproducibility 19**

# 1 Introduction and scope

Decoy-state BB84 with finite-key security needs five free parameters per operating point:

$$p = (\mu_1, \mu_2, P_{\mu_1}, P_{\mu_2}, q_x), \qquad \mu_3 = 2 \times 10^{-4} \text{ fixed}, \qquad P_{\mu_3} = 1 - P_{\mu_1} - P_{\mu_2}.$$

The optimal $p$ depends on the channel (fiber length $L$) and on the block size $n_X$. Re-optimising whenever the channel changes is the motivation for fast surrogates: neural networks, and next, quantum machine learning (QML).

A surrogate can be no better than (a) the physics model that defines $R$ and (b) the optimizer that produces its training labels. This report establishes both, and the yardstick against which a surrogate is scored.

**Document history.** This report supersedes three earlier documents (Appendix A):

- the MSc thesis report, May 2025 (`Project_Report/...Final_Report.pdf`);
- the results report, January/March 2026 (`result_report/results_report.tex`);
- the interview slides derived from that report.

Numbers in those documents should not be cited. Where they conflict with this report, this report holds.

**Contributions.**

1. A physics model (preset `lim`) that is audited against the paper equation by equation. It documents every choice the paper leaves open and is validated against an independent implementation (Section 2).
2. A penalty-free JAX optimizer with exact second-order steps and per-point certificates (Section 3).
3. A trusted reference and an evaluation protocol. The protocol defines regret, distance horizon, smoothness and timing, and requires independent cross-checks (Sections 4–5).
4. A protocol, fixed in advance, for benchmarking surrogate models, including QML (Section 7).

# 2 Model: finite-key decoy-state BB84 (Lim et al. 2014)

## 2.1 Channel, detection and error model

With attenuation $\alpha$, $\eta_{\text{ch}} = 10^{-\alpha L/10}$ and $\eta_{\text{sys}} = \eta_{\text{Bob}}\,\eta_{\text{ch}}$. For intensity $k \in \{\mu_1, \mu_2, \mu_3\}$ with probability $p_k$:

$$D_k = 1 - (1 - 2p_{\text{dc}})\, e^{-\eta_{\text{sys}} k}, \qquad R_k = D_k(1 + p_{\text{ap}}), \tag{1}$$

$$e_k = p_{\text{dc}} + e_{\text{mis}}\big(1 - e^{-\eta_{\text{sys}} k}\big) + \tfrac{p_{\text{ap}}}{2} D_k. \tag{2}$$

Both $1 - e^{-x}$ terms are evaluated as $-\text{expm1}(-x)$ (Section 2.7).

Expected counts are used as the observed counts (no sampling):

$$N = \frac{n_X}{q_x^2 \sum_k p_k R_k}, \qquad n_{X,k} = N q_x^2 p_k R_k, \quad n_{Z,k} = N(1 - q_x)^2 p_k R_k, \tag{3}$$

$$m_{X,k} = N q_x^2 p_k e_k, \quad m_{Z,k} = N(1 - q_x)^2 p_k e_k, \qquad e_{\text{obs}} = \textstyle\sum_k m_{X,k} / n_X. \tag{4}$$

## 2.2 Key length (Eqs. (1)–(5) of the paper)

Let $\delta(n,\varepsilon) = \sqrt{\frac{n}{2}\ln\frac{21}{\varepsilon}}$ and $n^{\pm}_{X,k} = \frac{e^k}{p_k}\big(n_{X,k} \pm \delta(n_X,\varepsilon_{\mathrm{sec}})\big)$. Define $n^{\pm}_{Z,k}$ likewise with $n_Z$, and $m^{\pm}_{Z,k}$ with $m_Z = \sum_k m_{Z,k}$. Let $\tau_j = \sum_k p_k e^{-k} k^j / j!$. Then

$$s_{X,0} \geq \tau_0 \,\frac{\mu_2 n^-_{X,\mu_3} - \mu_3 n^+_{X,\mu_2}}{\mu_2 - \mu_3}, \tag{5}$$

$$s_{X,1} \geq \frac{\tau_1\mu_1\big[n^-_{X,\mu_2} - n^+_{X,\mu_3} - \frac{\mu_2^2-\mu_3^2}{\mu_1^2}\big(n^+_{X,\mu_1} - \frac{s_{X,0}}{\tau_0}\big)\big]}{\mu_1(\mu_2-\mu_3) - \mu_2^2 + \mu_3^2}, \tag{6}$$

$$v_{Z,1} \leq \tau_1\,\frac{m^+_{Z,\mu_2} - m^-_{Z,\mu_3}}{\mu_2-\mu_3}, \qquad \phi_X \leq \frac{v_{Z,1}}{s_{Z,1}} + \gamma\Big(\varepsilon_{\mathrm{sec}}, \frac{v_{Z,1}}{s_{Z,1}}, s_{Z,1}, s_{X,1}\Big), \tag{7}$$

$$\gamma(a,b,c,d) = \sqrt{\frac{(c+d)(1-b)b}{cd\ln 2}\,\log_2\Big(\frac{c+d}{cd(1-b)b}\,\frac{21^2}{a^2}\Big)}, \tag{8}$$

$$\ell = s_{X,0} + s_{X,1}\big(1 - h(\phi_X)\big) - n_X f_{\mathrm{EC}} h(e_{\mathrm{obs}}) - 6\log_2\frac{21}{\varepsilon_{\mathrm{sec}}} - \log_2\frac{2}{\varepsilon_{\mathrm{cor}}}, \qquad R = \ell/N. \tag{9}$$

Here $h$ is the binary entropy. The secrecy parameter is tied to the key length, $\varepsilon_{\mathrm{sec}} = \kappa\ell$ with $\kappa = 10^{-15}$. We take the greatest self-consistent fixed point of $\varepsilon_{\mathrm{sec}} = \kappa\max(\ell(\varepsilon_{\mathrm{sec}}), 1)$, found by monotone iteration and log-bisection, and differentiate it exactly through the implicit-function theorem (`custom_jvp`).

## 2.3 Constants and presets

| Constant | value (`lim`) | note |
|---|---|---|
| $\alpha$, $\eta_{\mathrm{Bob}}$, $p_{\mathrm{dc}}$ | 0.2 dB/km, 0.1, $6\times10^{-7}$ | as in the paper |
| $e_{\mathrm{mis}}$, $p_{\mathrm{ap}}$ | $5\times10^{-3}$, $4\times10^{-2}$ | as in the paper |
| $f_{\mathrm{EC}}$, $\varepsilon_{\mathrm{cor}}$, $\kappa$ | 1.16, $10^{-15}$, $10^{-15}$ | as in the paper |
| $\mu_3$ | $2\times10^{-4}$ | fixed |
| box | $\mu_1 < 0.9$, $\mu_2 < 0.5$ | not stated in the paper; same box as all earlier runs |

Table 1: Constants of the default preset `lim` (`src/qkd_opt/presets.py`).

Three presets are used:

**`lim` (default)** Everything above. The misalignment term uses $\eta_{\mathrm{sys}}$, whereas the paper prints $\eta_{\mathrm{ch}}$.

**`lim_literal` (diagnostic)** The printed $\eta_{\mathrm{ch}}$ in $e_k$.

**`legacy`** The physics of the earlier code `src/qkd`: $e_{\mathrm{mis}} = 0.01$, $p_{\mathrm{ap}} = 0$, $\varepsilon_{\mathrm{sec}} = 10^{-10}$ fixed, $m_Z = \sum_k n_{X,k}$ (a defect), $R_k = D_k$, and $n^-, m^-$ clamped at 0. It is reproduced to $< 10^{-8}$ relative in $\ell$ and is used *only* to score the stored earlier runs on the objective they were optimised for.

## 2.4 Choices the paper leaves open

The paper does not print the following; each is our stated choice (full list in `PHYSICS_DIFF.md`):

- the expected-count model above;
- $e_{\mathrm{obs}}$ as the detection-weighted $X$-basis error rate;
- $\eta_{\mathrm{sys}}$ in the misalignment term;
- the greatest fixed point for $\varepsilon_{\mathrm{sec}} = \kappa\ell$;

- dropping the floor in Eq. (1), a continuous relaxation needed for gradients;
- clamping $s_{\cdot,0} \geq 0$ and dropping the single-photon term when $s_{\cdot,1} \leq 0$;
- capping $\phi_X \leq \frac{1}{2}$, and setting $\phi_X = \frac{1}{2}$ when $v_{Z,1}/s_{Z,1} \leq 0$;
- not clamping $n^-$, $m^-$;
- the factors $q_x^2$ and $(1-q_x)^2$ for basis sifting;
- the optimisation box.

### 2.5 Agreement with the paper

The paper states three numbers. No numerical constant was tuned to them. The one structural choice, $\eta_{\text{sys}}$ versus the printed $\eta_{\text{ch}}$, was made by comparing against these same numbers, so they do not independently validate that choice.

| Preset | $R(0, 10^9)$ [paper $\sim 10^{-2}$] | $R(100, 10^9)/R(100, 10^7)$ [$\sim 1.75$] | horizon $n_X = 10^4$ [$\sim 135$ km] |
|---|---|---|---|
| `lim` | $9.203 \times 10^{-3}$ (-8%) | 1.791 (+2%) | 130.4 (-3%) |
| `lim_literal` | $1.250 \times 10^{-3}$ (-87%) | 2.734 (+56%) | 19.0 (-86%) |
| `legacy` | $1.090 \times 10^{-2}$ (+9%) | 1.790 (+2%) | 148.2 (+10%) |

Table 2: Paper targets, with the relative deviation in brackets. The paper's values are read from its text and Fig. 1, so they are approximate ($\sim$). `lim` is within 8 %. The printed $\eta_{\text{ch}}$ misses all three by far, so we treat it as a probable typo. This is an inference; the authors have not confirmed it. `legacy` lands close only because its errors partly cancel (Table 3).

### 2.6 Effect of each correction

Table 3 applies the corrections from `legacy` to `lim` one at a time and re-optimises at each stage (`Optimization/analysis/physics_diff.py`; values† from `results/physics_diff_a419cc7422_dba94a5ecdc3/`).

| | stage (cumulative) | $R(100, 10^4)$ | $R(100, 10^9)$ | horizon $10^4$ | horizon $10^9$ |
|---|---|---|---|---|---|
| 0 | `legacy` | $2.60 \times 10^{-6}$ | $9.88 \times 10^{-5}$ | 148.17 | 190.67 |
| 1 | expm1 evaluation (numerics only) | $2.60 \times 10^{-6}$ | $9.88 \times 10^{-5}$ | 148.17 | 190.67 |
| 2 | $m_Z = Z$-basis errors (defect fixed) | $5.39 \times 10^{-6}$ | $1.42 \times 10^{-4}$ | 148.17 | 190.67 |
| 3 | $e_{\text{mis}}$ $0.01 \to 5 \times 10^{-3}$ | $8.20 \times 10^{-6}$ | $1.74 \times 10^{-4}$ | 152.50 | 192.71 |
| 4 | $p_{\text{ap}}$ $0 \to 0.04$ in $e_k$ | $1.22 \times 10^{-6}$ | $7.89 \times 10^{-5}$ | 133.61 | 182.43 |
| 5 | $R_k = D_k(1 + p_{\text{ap}})$ | $1.47 \times 10^{-6}$ | $8.57 \times 10^{-5}$ | 135.37 | 184.17 |
| 6 | $\varepsilon_{\text{sec}} = \kappa\ell$ | $1.07 \times 10^{-6}$ | $8.80 \times 10^{-5}$ | 130.44 | 184.16 |
| 7 | $n^-, m^-$ not clamped = `lim` | $9.91 \times 10^{-7}$ | $8.54 \times 10^{-5}$ | 130.44 | 184.16 |
| 8 | printed $\eta_{\text{ch}}$ = `lim_literal` | $\leq 0$ | $7.94 \times 10^{-6}$ | 18.97 | 149.31 |

Table 3: Cumulative ablation; $R$ in bits/pulse, horizons in km. **Reading:** (i) Fixing the $m_Z$ defect alone roughly doubles $R$ at $n_X = 10^4$. (ii) Afterpulsing (stages 4–5) shortens the $n_X = 10^4$ reach by 17 km. (iii) Overall, the earlier physics overstated the $n_X = 10^4$ reach by 17.7 km relative to `lim`.

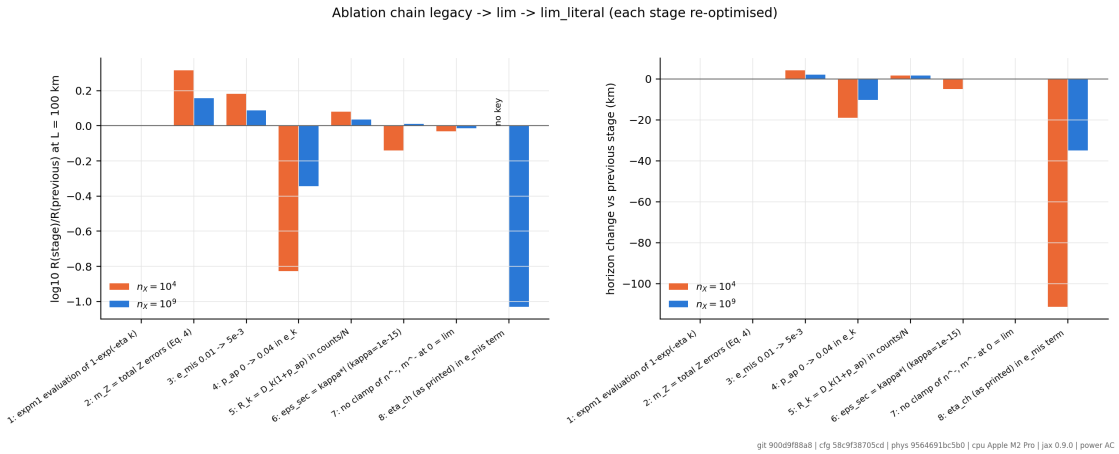



Figure 1: The ablation of Table 3 as $\Delta \log_{10} R$ at 100 km and Δhorizon (`fig7`).

### 2.7 Numerical correctness

**Independent implementation.** `tests/test_independent_physics.py` is a float64 and Decimal implementation written from the paper text that does not import the JAX code. With fixed $\varepsilon_{\text{sec}} = 10^{-10}$ and the `lim` switches it agrees on $\ell$ to $1.0 \times 10^{-13\dagger}$ at 360 interior points. With the full `lim` preset it agrees to $1.7 \times 10^{-13\dagger}$ at 18 near-horizon points, and on the $\varepsilon_{\text{sec}}$ fixed point to $6.3 \times 10^{-13\dagger}$. An 80-digit evaluation confirms that the error budget of Eq. (1) equals $21\varepsilon_{\text{sec}}$.

**Cancellation fix.** Near the horizon, $\eta_{\text{sys}} k \approx 10^{-8}$, and evaluating $1 - e^{-x}$ directly loses accuracy. Because $\ell$ is a small difference of terms up to 658 times larger, this produced a relative error in $\ell$ of up to $1.4 \times 10^{-8\dagger}$ against a 60-digit evaluation. With $-\text{expm1}(-x)$, a float64 implementation is within $\sim 6 \times 10^{-14\dagger}$ of the 60-digit value. The patched JAX preset was not re-measured against 60 digits. It agrees with that implementation to $1.7 \times 10^{-13\dagger}$. No horizon moved by more than $10^{-4}$ km.

**The closed-form $\gamma$ is conservative.** Lim's closed-form $\gamma$ exceeds the deviation $\theta$ obtained by solving the exact Fung–Ma–Chau tail bound for the same failure probability $a = \varepsilon_{\text{sec}}/21$, at every point checked (we did not prove this in general). At the 105 `lim` operating points the ratio is 1.28–1.44 (median 1.41)$^\dagger$. The exact form would extend the $n_X = 10^4$ horizon from 130.44 to 132.35 km$^\dagger$, and moves it by at most 0.21 km$^\dagger$ for $n_X \geq 10^6$. It is kept as a diagnostic preset (`lim_tight`), not as the default (Fig. 9).

## 3 Optimizer

**Feasibility by construction.** $\theta \in \mathbb{R}^5 \mapsto p$ through sigmoids and nested affine maps (`reparam.py`). These guarantee

$$\mu_3 < \mu_2 < 0.5, \qquad \mu_2 + \mu_3 < \mu_1 < 0.9, \qquad P_{\mu_k} > 0,\ \textstyle\sum_k P_{\mu_k} = 1, \qquad 0 < q_x < 1.$$

There are no penalty terms. The earlier code used a $10^{250}$ step penalty, which has zero gradient.

**Objective.** Where a positive key exists, the optimizer maximises $R$. Beyond the horizon, maximising $R$ is degenerate: $R < 0$ is pushed towards $0^-$ by $q_x \to 0$. There the optimizer maximises $\ell/n_X$ instead, which is smooth and has the same sign as $\max_p R$ (because $N > 0$). The horizon is then exactly the zero of $\max_p \ell$. Neither $R$ nor $\ell$ is ever clipped. The objective is divided by its value at the best start, so tolerances are relative ($R$ spans $10^{-2}$ to $10^{-11}$).

**Global and local stages.**

- *Global:* scrambled Sobol points in $u$-space (mapped to $\theta$), evaluated with `jit`+`vmap`; the diverse top-$k$ are kept.
- *Local:* saddle-free damped Newton with the exact $5 \times 5$ Hessian (absolute eigenvalues, regularised), a step cap and Armijo backtracking, all in float64.
- *Warm sweep:* continuation along $L$. A global re-check runs every 10 points. A branch switch found by a re-check is logged, and the sweep backfills backwards from it. Changes between re-checks that need no switch are not logged (open issue 1). the horizon is refined to $\ell = 0$ by bisection.

**Certificate.** Each point is returned with:

- a status, e.g. `converged_grad` or `no_descent`;
- $\|\nabla_\theta\|_\infty$ of the scaled objective (objective divided by |value at the best start|);
- the Hessian eigenvalues;
- the active bounds;
- a branch id.

# 4 Evaluation protocol

## 4.1 Trusted reference

**Grids.**

- *Bulk:* 21 fiber lengths (every 20th point of `linspace(0,200,400)` plus the last; about 10 km apart; the same $L$ values as the earlier runs) $\times$ $n_X \in \{10^4, \ldots, 10^9\}$, i.e. 126 points.
- *Tail:* every 0.501 km grid point from 15 km below to 1 km above each horizon, i.e. 192 points.

**Construction.** For `lim` and `lim_literal`, each reference point is the best of four candidates:

- two independent cold solves ($2^{17}$ Sobol points, seeds 0/1 bulk and 2/3 tail, top 64 refined by Newton);
- two neighbour-continuation solves, one from each side in $L$.

The `legacy` reference uses the two Sobol solves plus Newton refinement started from the three stored earlier runs. It was not rebuilt with continuation candidates.

**Acceptance tests.**

1. The two seeds agree to $3.3 \times 10^{-13\dagger}$ (bulk) and $7.6 \times 10^{-13\dagger}$ (tail).
2. An *independent* optimizer (SciPy SLSQP in $p$-space, random starts, different code path) never beats the reference by more than the tolerance $\mathrm{tol}_{\mathrm{ref}} = \max(10^{-9}, 2 \times \text{seed disagreement})$: 0 weaknesses in the bulk and 0 in the tail.
3. $10^5$ random perturbations (scales $10^{-6}$–$10^{-1}$) around each of 72 tail points improve $R$ by at most $4.9 \times 10^{-13\dagger}$.

SLSQP itself is weaker than the reference near the horizon: it agrees within $10^{-6}$ at 98.1 % (bulk) and 87.8 % (tail) of points, and is otherwise *worse*. That makes it a test the reference can fail, not one it is guaranteed to pass.

**cliff_edge points.** A point with $R > 0$ and stored certificate gradient $> 10^{-3}$ is labelled *cliff_edge*. Such points are excluded from scores and listed separately: 2 of 105 bulk points with $R > 0$, and 5 of 180 tail points†.

**What the flag does and does not mean.** The flag does *not* mark an unconverged optimum or a discontinuity of $R(p)$.

- *Why the stored gradient is large.* It is the gradient of the objective divided by the objective's value at the solver's best *starting* point. At all 7 points, which lie within 1.2 km of their horizon, that starting point had a near-zero key rate (about $10^{-28}$–$10^{-26}$). $R$ at the solution is $10^{-12}$–$10^{-10}$, so the stored gradient is inflated by a factor of about $10^{15}$–$10^{17}$.
- *What the points really are.* Normalised by $R$ at the solution instead, their gradient is $2.8 \times 10^{-13}$–$1.1 \times 10^{-6}$†. They are converged optima.
- *Why they are still excluded.* The stored certificate cannot confirm them near the horizon, so they stay out of the scores and the counts above are unchanged.

The diagnosis is in `STATUS.md` §10a on branch `qml-surrogate`: re-runs of the winning solves reproduce $R$ to 11 digits.

Points with gradient between $10^{-6}$ and $10^{-3}$ (*noise_floor*) are scored. The same normalisation applies to that tier. This applies to `lim` and `lim_literal`. The `legacy` references were built before this rule and use the earlier threshold, $10^{-6}$. Their large gradients also arise as $|R| \to 0$ at the horizon; the same normalisation is the likely cause, but this has not been re-checked.

### 4.2 Metrics

**Regret** $r = (R_{\text{ref}} - R(\hat{p}))/R_{\text{ref}}$, with $R(\hat{p})$ re-evaluated by the float64 model at the method's parameters. We report the median, p99, maximum, and the fraction with $r \leq 10^{-6}$ (also $10^{-4}, 10^{-3}, 10^{-2}$). $r \geq 1$ means the parameters give no key.

**Horizon** The largest $L$ with $\max_p \ell > 0$, found by bisection. It is *not* a threshold on $R$; the earlier $R \geq 10^{-12}$ convention was arbitrary.

**Smoothness** Total variation per km of the box-normalised parameters on $S = \{L : R_{\text{ref}}(L) > 10^{-3} R_{\text{ref}}(0)\}$. Logged branch-switch pairs are excluded and listed. Restricting to $S$ removes the region beyond the horizon, which dominated the earlier roughness numbers.

**Timing** Wall-clock time on one machine (Apple M2 Pro, CPU backend, float64, JAX 0.9.0, AC power, no other jobs). JIT compile time is reported separately. Device-resident and end-to-end medians are both given, and every timing names its batch size and warm or cold start.

## 5 Results

### 5.1 Horizons and key rate

| Preset | $10^4$ | $10^5$ | $10^6$ | $10^7$ | $10^8$ | $10^9$ |
|---|---|---|---|---|---|---|
| `lim` | 130.44 | 160.61 | 172.62 | 178.67 | 182.09 | 184.16 |
| `lim_literal` | 18.97 | 104.53 | 129.48 | 140.00 | 145.85 | 149.31 |
| `legacy` | 148.17 | 171.43 | 180.97 | 185.97 | 188.87 | 190.67 |

Table 4: Reference horizons (km) for each $n_X$. The new warm sweep reproduces the `lim` horizons to within 0.0072 km†.

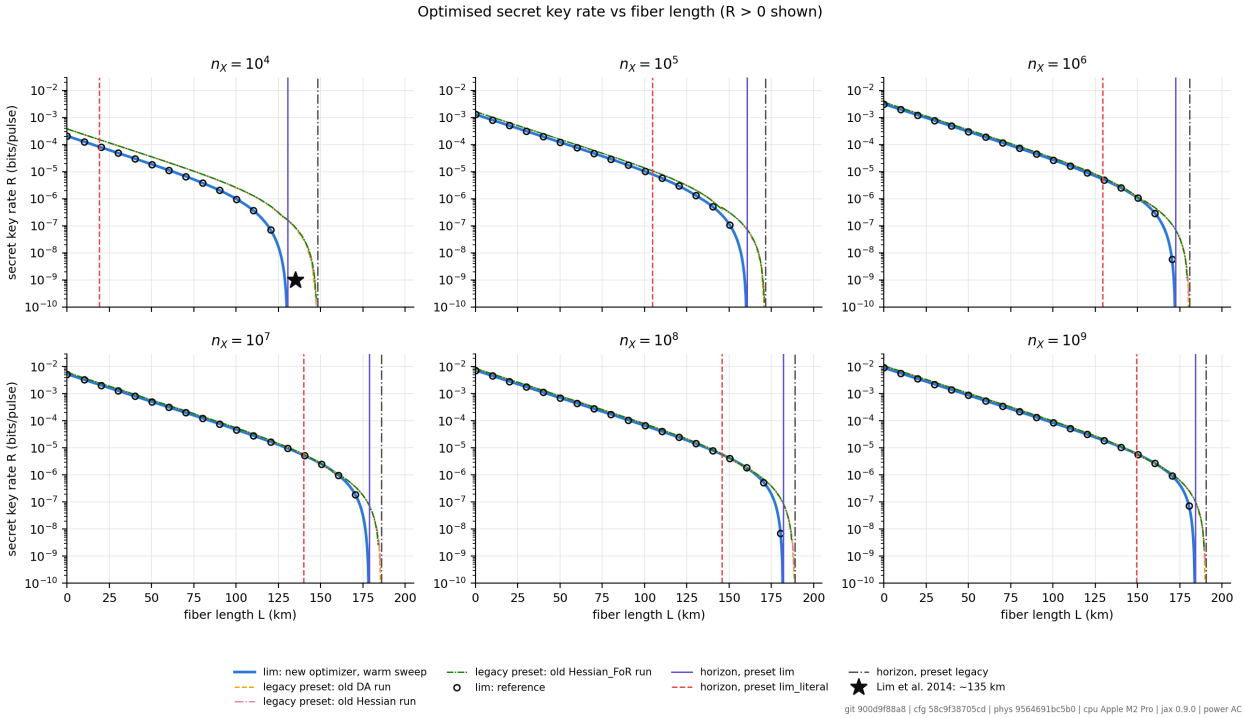



Figure 2: Optimal key rate under `lim` (warm sweep and reference points), with the stored earlier runs evaluated under `legacy`, the objective they were optimised for (`fig1`).

## 5.2 Accuracy of the optimal parameters and key rates

| Preset | Method | % $r \le 10^{-6}$ (bulk / tail) | median $r$ (bulk) | max $r$ (bulk / tail) |
|---|---|---|---|---|
| `lim` | new, cold start | 100.0 / 100.0 | $2.37 \times 10^{-15}$ | $9.06 \times 10^{-14}$ / $6.47 \times 10^{-13}$ |
| `lim` | new, warm sweep | 100.0 / 100.0 | $3.81 \times 10^{-15}$ | $1.03 \times 10^{-13}$ / $9.50 \times 10^{-13}$ |
| `lim` | lookup table (2 km nodes) | 81.6 / 24.6 | $1.36 \times 10^{-9}$ | $2.17 \times 10^{-2}$ / $2.31 \times 10^{1}$ |
| `legacy` | new, warm sweep | 100.0 / 100.0 | $5.87 \times 10^{-13}$ | $1.40 \times 10^{-9}$ / $2.61 \times 10^{-9}$ |
| `legacy` | earlier DA run | 3.7 / 0.0 | $1.42 \times 10^{-2}$ | $1.26 \times 10^{0}$ / $8.91 \times 10^{-1}$ |
| `legacy` | earlier Hessian run | 96.3 / 95.8 | $1.14 \times 10^{-11}$ | $2.68 \times 10^{1}$ / $4.67 \times 10^{-1}$ |
| `legacy` | earlier Hessian_FoR run (NN labels) | 81.7 / 0.0 | $2.03 \times 10^{-11}$ | $9.65 \times 10^{1}$ / $3.64 \times 10^{1}$ |

Table 5: Regret against the trusted reference, with cliff_edge points excluded (bulk: 103 `lim` / 109 `legacy` points; tail: 175 / 142; the `legacy` counts use the earlier cliff_edge threshold)†. The earlier runs are scored on their own objective (`legacy`). The cold start uses Sobol seed 7, independent of the reference seeds. The lookup table interpolates the new optimizer's own optima linearly in $L$ between nodes 2 km apart, separately for each $n_X$; it is the simplest surrogate and the floor for Section 7. Regret above 1 means the parameters yield no key.

**Reading.** On `lim`, the new optimizer is at the reference to $10^{-12}$ or better everywhere, including the near-horizon tail, where the earlier DA and Hessian_FoR runs were within $10^{-6}$ at 0 % of points. The run that produced the earlier neural-network labels missed the optimum by more than $10^{-6}$ at 18 % of bulk points and at every tail point. Its worst bulk regret of 96 means a negative key rate. Any surrogate trained on those labels inherits these errors.

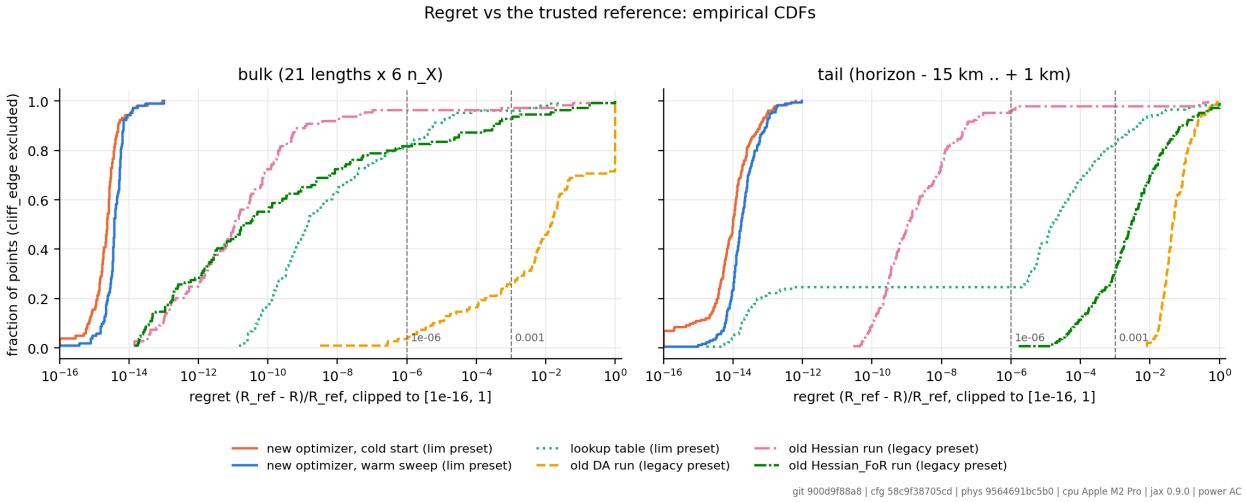



Figure 3: Empirical CDF of regret, bulk and tail (`fig6`).

### 5.3 Structure of the optimal parameters

**Branch switches.** The stored earlier runs jump repeatedly between 148 and 157 km at $n_X = 10^{9\dagger}$:

- DA: 5 jumps;
- Hessian_FoR: 5 jumps;
- Hessian: 1 jump.

For `lim`, the optimal parameters also switch between competing local optima while the key is positive. At such a crossover the parameters jump while $R$ is continuous. At the six reference block sizes, at least ten such switches were found$^{\dagger}$. Each is confirmed by independent cold solves ($2^{14}$ Sobol, top-16, 0.05 km grid, where one parameter step is 35–1,522× the median step and $|\Delta R|/R$ equals its neighbours), by the same jump test on the 1/32 km label lattice, or (the second $10^5$ switch) by two converged optima at the same lattice points, the backward sweep's being better by up to $6.2 \times 10^{-5}$ (`STATUS.md` §12):

| $n_X$ | crossover (km) | found by |
|---|---|---|
| $10^5$ | 130.5–130.6 | benchmark sweep (detected at 135.3 km, 7.7 % gain); dense forward and backward sweeps |
| $10^5$ | 132.72–132.75 | backward sweep over the label lattice; the dense forward sweep switched about 0.09 km late (132.81–132.84) |
| $10^6$ | 118.5–118.6 | benchmark sweep (detected at 120.3 km, 0.6 % gain); dense forward and backward sweeps |
| $10^6$ | 138.50–138.55 | dense forward sweep (unlogged jump) |
| $10^7$ | 105.35–105.40 | backward sweep; the dense forward sweep placed it 0.55 km too late (105.95) |
| $10^7$ | 138.20–138.25 | dense forward and backward sweeps |
| $10^8$ | ≤91.85 | dense forward sweep (unlogged jump at 92.4) |
| $10^8$ | 134.75–134.80 | dense forward sweep |
| $10^9$ | 77.55–77.60 | dense forward sweep (unlogged jump at 77.75; open issue 1) |
| $10^9$ | 128.65–128.70 | dense forward sweep |

Table 6: Below-horizon switches between competing optima at the six reference block sizes. None was found at $n_X = 10^4$.

The list is not exhaustive. Cold-solve confirmation was run only around candidate jumps, and a cold solve ($2^{14}$ Sobol, top-16) can itself miss the better branch next to a crossover: at 21 label points a warm sweep (forward or backward) beats it by up to $3.7 \times 10^{-4}$ (`STATUS.md` §12). Cold-solve agreement is therefore a lower-bound check, not proof of global optimality. An earlier version did not count the $10^5$ switch at 132.7 km, because cold solves showed only an 8× step there.

Sources: benchmark sweep `results/benchmark_lim_*/summary.json`; dense forward label sweep on branch `qml-surrogate` (`STATUS.md` §10, §10a, §10d); backward sweep (§11(b)); cold solves (§11(c)); label repair and the cold-solve limit (§12).

The benchmark warm sweep logged only the first two switches. At the reference points it is nevertheless optimal: it scores 100 % in Table 5, and an independent backward sweep agrees with the reference there (`STATUS.md` §11(b)). Between reference points, a forward warm sweep can stay on the worse branch past a crossover for up to about 0.6 km (`STATUS.md` §10d, §11(c)).

A dense 1 km scan flags parameter changes larger than 0.05 (normalised) only in the last ∼5 km before each horizon. Refined to 0.1 km, the largest step is 11–12 % of the 1 km change below the horizon, i.e. continuous drift. For pairs that cross the horizon it is 16–54 %. The dense scan did not flag the switches above. Their relation to the scan's threshold and normalisation has not been investigated (open issue 6).

**Smoothness.** TV/km on $S$, with logged switches excluded:

- `lim` warm sweep: $2.38 \times 10^{-3}$ ($n_X = 10^4$) and $6.54 \times 10^{-4}$ ($10^9$).
- `legacy`, $n_X = 10^4$: new warm sweep $1.85 \times 10^{-3}$; earlier DA $7.95 \times 10^{-3}$ (4.3× rougher); earlier Hessian_FoR $1.84 \times 10^{-3}$; earlier Hessian $1.85 \times 10^{-3\dagger}$.
- `legacy`, $n_X = 10^9$: new warm sweep $1.86 \times 10^{-3}$; earlier DA $2.61 \times 10^{-3}$ (1.4× rougher); earlier Hessian_FoR $1.85 \times 10^{-3}$; earlier Hessian $1.84 \times 10^{-3\dagger}$.

On the region with a useful key, the new optimizer is as smooth as the earlier Hessian-based runs and smoother than DA. Smoothness is therefore not where it improves. The improvement is optimality: Table 5, especially in the tail. The earlier "129× stability" figure came from the region beyond the horizon and is withdrawn.

**Note on comparing parameter values.** In Fig. 4 the `lim` curves and the earlier runs optimise different physics, so their parameter *values* are not comparable. Only continuity is compared.

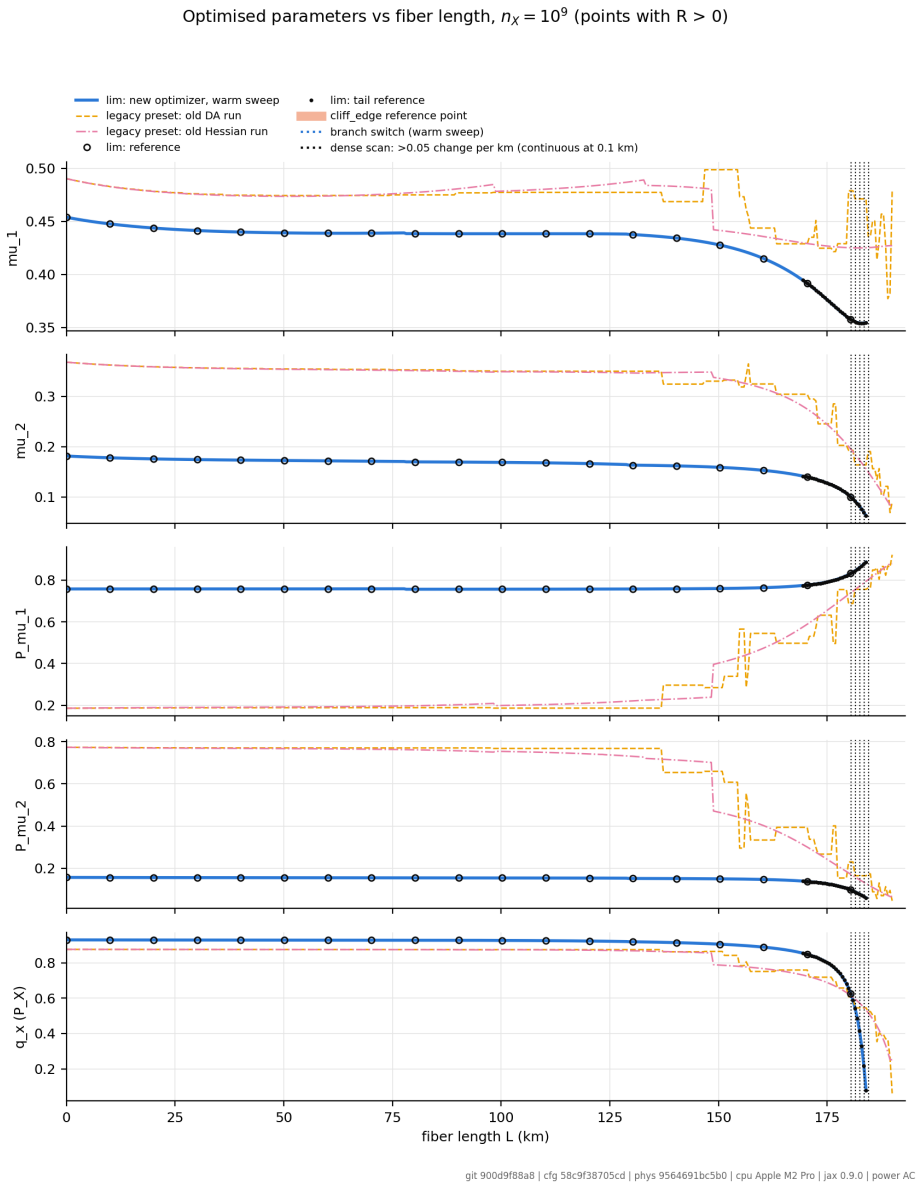



Figure 4: Optimal parameters vs $L$ at $n_X = 10^9$ (`fig4`; the $n_X = 10^4$ panel is `fig4_params_nx1e4.png`).

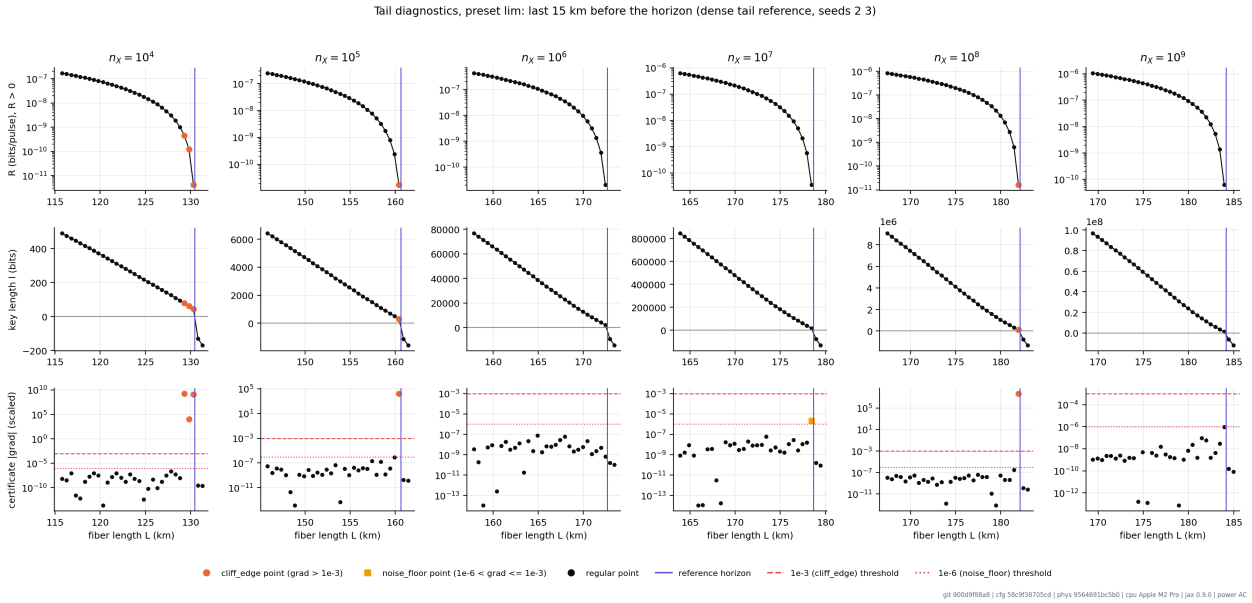



Figure 5: The last 15 km before each horizon: $R$, $\ell$ and the certificate gradient. Orange: cliff_edge; yellow: noise_floor (`fig5`).

## 5.4 Computational cost

| Method | start | batch | time per point |
|---|---|---|---|
| Earlier pipeline, $L \leq 100$ km branch (L-BFGS-B) | cold | 1 | 0.7–10 ms$^\dagger$ |
| Earlier pipeline, $L > 100$ km branch (DA, `maxiter` 5000–15000, 50k–150k evaluations) | cold | 1 | 5.8–17.6 s |
| New optimizer, $2^{12}$ Sobol + Newton | cold | 126 (vectorised) | 40.0 ms$^\dagger$ (compile 38.4 s$^\dagger$) |
| New optimizer, full sweep of 2400 points | warm | 6 chains | 4.27 ms (compile 80.9 s) |
| New optimizer, 100 km, $n_X = 10^9$ | warm (0.5 km away) | 1 | 1199 $\mu$s device / 1466 $\mu$s end-to-end |
| New optimizer, 5 km below each horizon | warm | 1 | 1.7–2.6 ms$^\dagger$ device |
| Lookup table (timing variant: 2.5 km $\times$ log $n_X$ grid) | – | 1 | 7.7 $\mu$s device / 10.0 $\mu$s end-to-end |

Table 7: Measured wall time on the `lim` objective, single machine (`results/timing_lim_*`, `results/benchmark_lim_*`).

**Fair comparisons.**

- **Cold vs cold:** a DA call (one point) against the new cold solver (amortised over a vectorised batch) is a factor of about 150–440. A batch-1 cold solve was not timed.
- **Warm solve vs the earlier pipeline at the same distances.** Below 100 km the earlier pipeline used L-BFGS-B (3.9 ms$^\dagger$ at 100 km, $n_X = 10^9$), only about 3$\times$ slower than the warm solve, and not optimal. Above 100 km it used DA. At the timed DA points where the reference has a positive key (110 and 150 km at $n_X = 10^9$; 130 km at $10^4$, where DA itself returned $R \leq 0$; 5.8–11.7 s$^\dagger$), a warm solve (1.2–2.6 ms) is about $2 \times 10^3$–$10^4$ times faster. The 17.6 s DA point (190 km) lies beyond the horizon. The warm solve uses a solution 0.5 km away, prior information that DA did not have.
- **Quality:** the new solver is within $10^{-6}$ at 100 % of points, against 3.7 % for the earlier DA run.
- The lookup-table latency is measured on a slightly different table (2.5 km nodes, interpolated also in $\log n_X$) from the one scored in Table 5. Both are $O(10)$ $\mu$s.

These ratios replace the withdrawn $5.9 \times 10^6$, 3,574$\times$ and 270$\times$ figures.

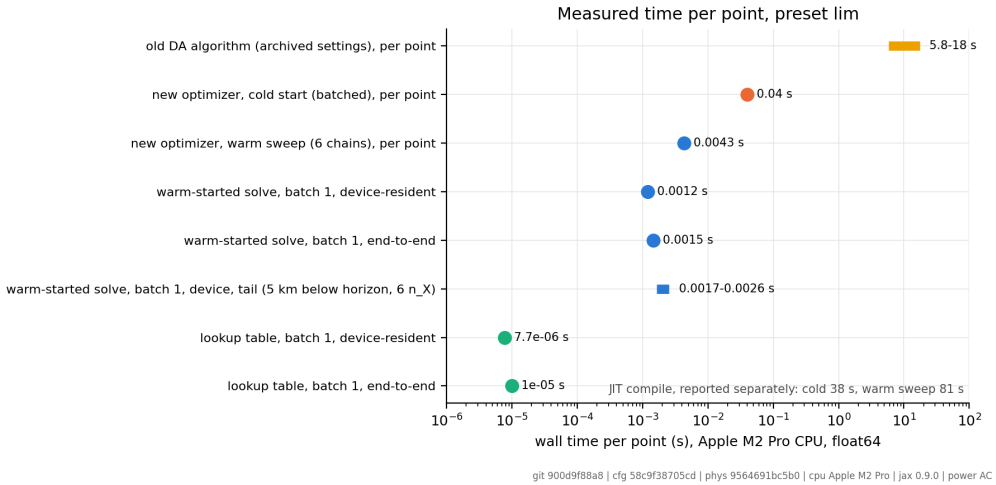



Figure 6: Measured wall time per point (`fig8`).

# 6 Discussion

## 6.1 What changed relative to the earlier work

1. **The model.** The earlier key rate contained a defect ($m_Z$) and departed from the paper in four settings (Table 3). Its $n_X = 10^4$ horizon (148.2 km) was 17.7 km longer than that of the corrected model (130.4 km). The corrected model's horizon is 3 % below the paper's ∼135 km.

2. **The labels.** Optimal parameters are now certified, not assumed. The earlier labels were sub-optimal at 18 % of bulk points and at all tail points, so the earlier NN accuracy figures measured agreement with sub-optimal targets.

3. **The metrics.** Regret on re-evaluated $R$ replaces relative parameter error. The horizon is $\ell = 0$ instead of $R \geq 10^{-12}$. Smoothness excludes the no-key region. Timings compare equal work and state batch size and warm or cold start.

## 6.2 Threats to validity

1. **Model fidelity.** `lim` matches the paper to within 8 %, not exactly, and rests on the choices in Section 2.4. Using $\eta_{\text{sys}}$ instead of the printed $\eta_{\text{ch}}$ is an inference from Table 2. All results are conditional on this model. Changing it changes the numbers but not the method.

2. **Expected counts.** Observed counts are set to their expectations (a choice the paper does not print), so statistical fluctuations of an actual run are not modelled.

3. **Continuous relaxation.** Dropping the floor in Eq. (1) changes $\ell$ by less than one bit, i.e. relatively by at most $1/\ell$. That is negligible in the bulk and about 2 % at the smallest cliff_edge keys ($\ell \approx 45$ bits).

4. **The reference is the best known optimum, not a proof of global optimality.** The evidence for it is:

   - two independent seeds;
   - an independent optimizer (SLSQP) on a different code path;
   - a local perturbation test;
   - an independent backward warm sweep on a dense descending grid (1,999 fiber lengths). It agrees with the reference to about $10^{-12}$ at all 285 points with $R > 0$, and never beats it by more than $10^{-9}$ (`STATUS.md` §11(b)). Its limitation: the bulk reference grid is about

10 km apart, so it does not sample the sub-km stretches where a forward sweep was found to stay on a worse branch.

The reference and the new optimizer share the physics and the Newton solver. The physics is checked independently (Section 2.7); the SLSQP cross-check is the independent check on the optimisation.

5. **Excluded points.** The cliff_edge exclusions (2 bulk, 5 tail) remove 7 points near the horizon whose stored certificate gradient is inflated by its normalisation (Section 4). Re-checked, they are converged optima, not discontinuities. Scores there are listed separately, not hidden.
6. **The earlier runs scored here** are the February 2026 script outputs. The 2025 thesis dataset and network were not re-scored (Appendix A).
7. **Timing** comes from one CPU machine; GPU and other hardware were not measured.

### 6.3 Open issues

None of these affects the `lim` results above (from `STATUS.md`):

1. A $\sim$0.002 (normalised) parameter step at $n_X = 10^9$, 77.7–78.2 km, is not logged as an event; $R$ is smooth there. Proposed fix: a `drift_step` log.
2. `lim_literal` tail (diagnostic): the new optimizer misses the reference by up to $8.74 \times 10^{-2}$ at 3 of 146 points. Not yet investigated.
3. `legacy` tail: one cold-start point has regret $1.14 \times 10^{-1}$. The `legacy` references were not rebuilt with continuation candidates.
4. The sensitivity, robustness and extrapolation analyses (`analyses.py`) have not been re-run since the expm1 fix.
5. The neural network has not been retrained (`NN_RETRAIN_PLAN.md`).
6. The below-horizon branch switches (Table 6) were not flagged by the dense scan, and no single sweep logs all of them. Characterise the competing optima, and make switch detection reliable at this size, for example with a backward pass.

## 7 Benchmark protocol for surrogates (classical NN and QML)

The value of a surrogate is a trade-off between *regret* and *latency*. The protocol below is fixed *before* any model is trained, so that the classical and quantum models are compared on identical terms. It encodes the lessons of the earlier work.

**Task.** Learn $x \mapsto p^\star(x)$ with $x = (L, \log_{10} n_X)$, optionally extended by $p_{\mathrm{dc}}$, $e_{\mathrm{mis}}$ and $\alpha$ (ranges in `NN_RETRAIN_PLAN.md`). Models predict in $\theta$- or $u$-space, so that every prediction is feasible by construction.

**Labels.**

- Generated where compute is cheap (e.g. cloud CPU/GPU), not on a laptop.
- Generated by the new optimizer under `lim` (or under the extended model, with the same verification).
- Each label carries its certificate.
- cliff_edge points are excluded from training and from scoring, and listed. Near the two crossovers (118.5 and 130.5 km), $p^\star$ is discontinuous. Report them separately.

**Splits (no test point on the training grid).**

- *Interpolation:* held-out $L$ between training nodes, and held-out block sizes (e.g. $n_X = 5 \times 10^8$, as in the thesis).
- *Tail:* the dense tail grid, scored separately.
- *Extrapolation:* reported separately, never pooled with interpolation.

**Primary metric.** Regret of the float64 $R$ re-evaluated at the predicted parameters, reported for bulk and tail separately:

- the median, p99 and maximum;
- the fractions with $r \leq 10^{-6}$, $10^{-4}$, $10^{-3}$ and $10^{-2}$;
- the fraction with no key ($r \geq 1$).

Secondary metrics: horizon error, and parameter error in $u$-space. Parameter error alone is not sufficient because $R$ is flat in some directions and steep in others.

**Baselines.** A surrogate is compared against three references:

1. the lookup table: 81.6 % (bulk) and 24.6 % (tail) within $10^{-6}$, at 10.0 $\mu$s;
2. a classical NN with a matched parameter count;
3. the warm-started exact solver: 100 % within $10^{-6}$, at 1466 $\mu$s.

We also report the hybrid "surrogate prediction + $k$ Newton steps" for $k = 1, 2, 3$. Whether it beats both components has not been measured.

**Cost accounting.**

- Latency: batch-1 end-to-end and device-resident, plus batched throughput.
- Training cost.
- For QML: the backend (simulator or hardware), the shots per prediction and the circuit evaluations per prediction. On hardware these dominate latency and must not be omitted.

**Statistics.** At least 5 training seeds, reported as mean $\pm$ standard deviation. Hyperparameters are chosen on a validation split, never on the test split.

**Usefulness criterion.** A surrogate is useful if either of the following holds:

- it reaches the warm solver's regret class at lower latency;
- it beats the lookup table's regret at comparable latency.

Every comparison in the QML project should be framed against this criterion.

# 8 Conclusion

The parameter-optimisation problem for finite-key decoy-state BB84 is now solved to within $10^{-12}$ relative regret of an independently cross-checked best-known reference (cliff_edge points excluded), on a corrected model which is validated against the paper and against an independent implementation:

- the optimal parameters vary smoothly with distance, apart from switches between competing optima (at least ten below the horizon at the six block sizes);
- the horizons and key rates are reproducible from scripts with recorded provenance;
- the earlier speed-up, stability and neural-network claims are withdrawn.

The reference, metrics and protocol of Sections 4 and 7 are the basis for the classical-versus-quantum surrogate benchmark.

# A Earlier documents and their status

| Document | Setup | Status |
|---|---|---|
| MSc thesis report, May 2025 (`Project_Report/`) | $p_{\mathrm{dc}} = 6 \times 10^{-7}$, $e_{\mathrm{mis}} = 5 \times 10^{-3}$, $p_{\mathrm{ap}} = 0$ (stated), $\varepsilon_{\mathrm{sec}} = 10^{-10}$ fixed; 6,000 DA labels; NN with 4 inputs and 5 outputs | **Superseded.** Its own labels and NN are not re-scored; its DA pipeline was re-run under `lim` and scored (below).<br>• Its physics omits afterpulsing and uses a fixed $\varepsilon_{\mathrm{sec}}$. In Table 3, these two differences (stages 4–6) shorten the $n_X = 10^4$ horizon by 22 km.<br>• Its horizons (∼145–180 km) use an $R \approx 10^{-8}$ threshold, not $\ell = 0$.<br>• The DA pipeline was re-run on the thesis grid (1,000 distances in $[0, 200]$ km × $n_X = 10^4, \ldots, 10^9$; 6,000 points) under `lim` physics, because the thesis physics is not a defined preset. The runs used CPU cores of CINECA Leonardo, and each point was scored against the verified optimizer ($2^{12}$ Sobol, top-8) at the same point (`results/da_dataset_lim/DA_DATASET.md`; `STATUS.md` §13). Where the optimizer finds a key (5,041 points), the median regret is $5.0 \times 10^{-2\dagger}$, and 8.7 %† are within $10^{-6}$.<br>• $L \leq 100$ km, L-BFGS-B from the archived default guess: no key at 66.7 % (2,000 of 3,000)† of the points where a key exists. The failure is systematic. For $n_X = 10^4$–$10^7$, every solve stops after about 5 evaluations with $P_{\mu_1}$ at its $10^{-12}$ bound. For $10^8$ and $10^9$ the median regret is $1.4 \times 10^{-6}$ and $5.9 \times 10^{-6\dagger}$.<br>• $L > 100$ km, dual annealing: median regret $2.7 \times 10^{-2\dagger}$; no key at 1.0 % (21 of 2,041)†.<br>• Compute: 31.3 CPU-hours† (median 36 s per dual-annealing point).<br>• The 3.7 % figure in Table 5 belongs to a different DA run (February 2026, `legacy` physics), not to the thesis labels.<br>• "±6 % SKR error" and "270×" are not reproduced and should not be cited. The 270× divides an estimated DA time (4.5 min for 100 points, scaled from 45 min per 1,000) by a measured NN time (1 s).<br>• The background chapters (QKD, optimisation, NNs) remain valid as background. |
| Results report, Jan./Mar. 2026 (`result_report/superseded/results_report_2026-03.pdf`) | `legacy` physics; Hessian_FoR labels; NN with mismatched checkpoints | **Withdrawn claims.** All 40 numeric claims are classified in `CLAIMS.md`; none of the headline numbers ($5.9 \times 10^6$, 11,000×, 3,574×, 129×, "20 km+", "16 %") is reproducible or well defined. Its NN cannot be re-evaluated. The current 6-output checkpoint does not load into the 5-output network the evaluation scripts build, and no record says which scaler pair belongs to which checkpoint (`results/nn_legacy_check/`). |
| Interview slides (`result_report/superseded/`) | same as the results report | **Withdrawn claims**, as for the results report (same numbers). |

Table 8: Status of earlier documents. None of them is deleted; they are kept as historical record.

# B Additional figures

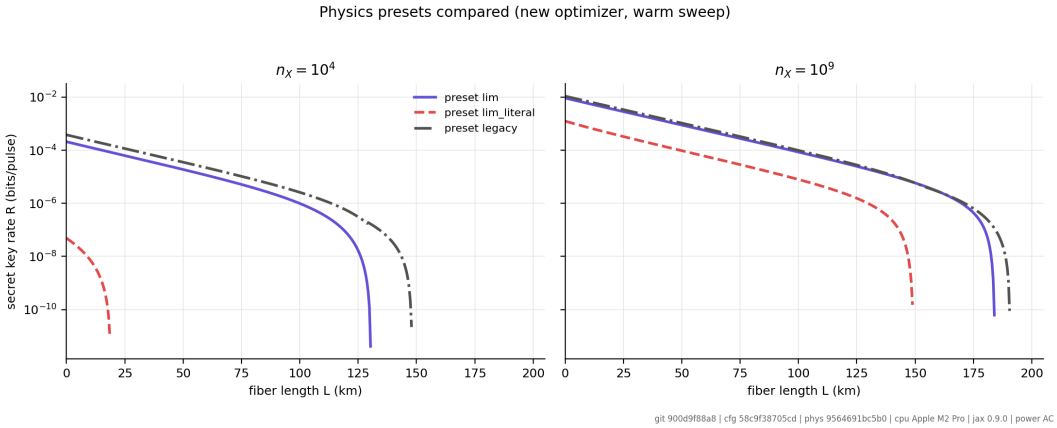



Figure 7: $R(L)$ under the three presets for $n_X = 10^4$ and $10^9$ (`fig2`).

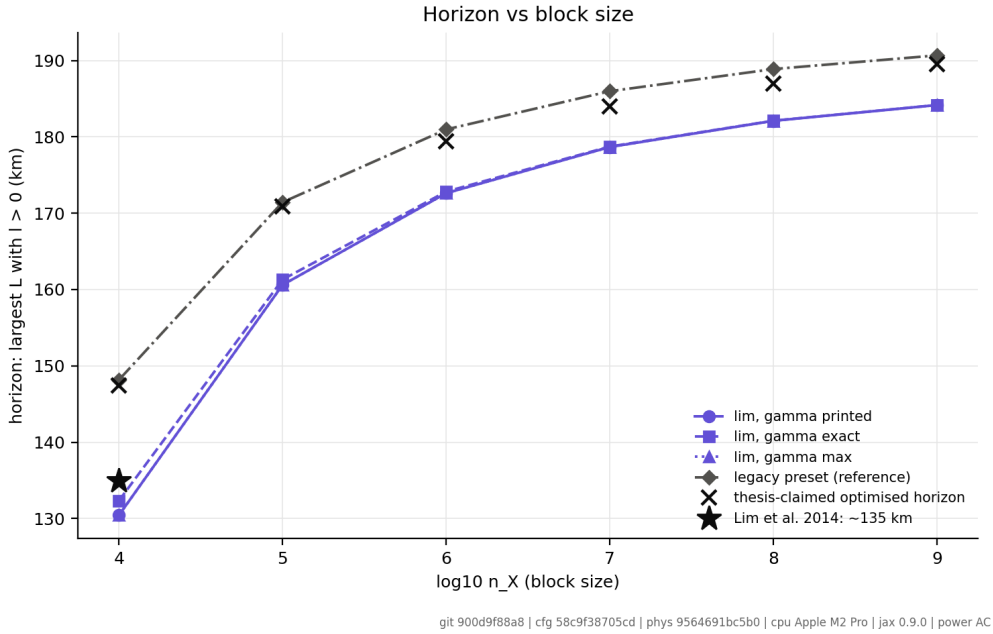



Figure 8: Horizon vs $n_X$: `lim` with the printed, exact and max $\gamma$; `legacy`; the horizons claimed in the earlier thesis material; the paper's $\sim$135 km (`fig3`).

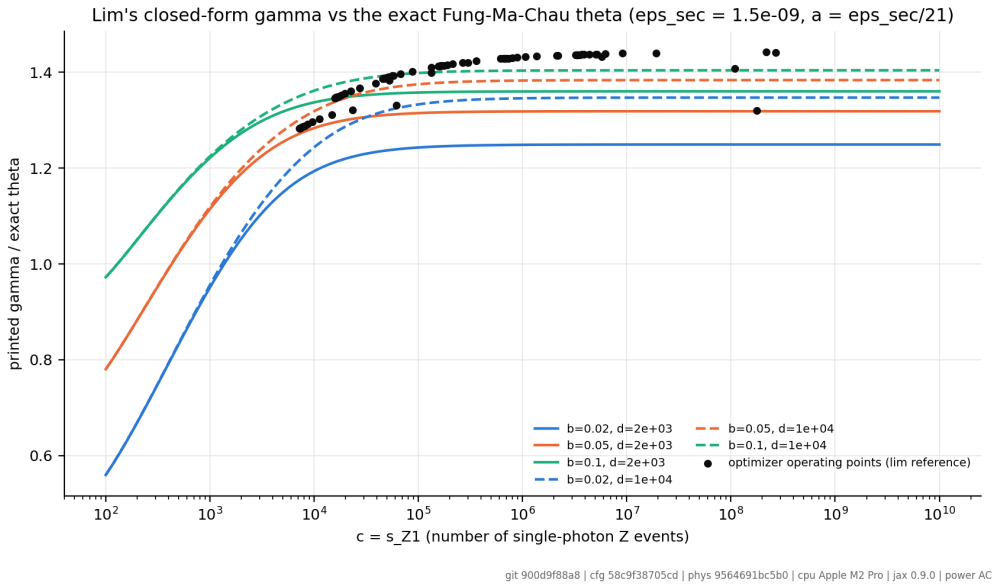



Figure 9: Ratio of the closed-form $\gamma$ to the exact Fung–Ma–Chau deviation; black points are the operating points (`fig9`).

# C Reproducibility

- `python Optimization/run_all.py` rebuilds the references, cross-checks, benchmarks, timings and analyses. Each step is recorded in `results/MANIFEST.json` with git commit, config hash, physics hash, hardware and power state.
- `Optimization/report/make_numbers.py` writes `generated_numbers.tex`, which defines every `\QKD...` value used here.
- `Optimization/report/make_figures.py` writes the figures (git `900d9f88a8`, config `58c9f38705cd`).
- Numbers marked † are copied from the result file named beside them.
- The test suite passes: 34 tests, `python -m unittest discover tests`.
- The run log with every measured value is in `STATUS.md`.

# References

1. C. C. W. Lim, M. Curty, N. Walenta, F. Xu, H. Zbinden, *Concise security bounds for practical decoy-state quantum key distribution*, Phys. Rev. A **89**, 022307 (2014), arXiv:1311.7129.
2. C.-H. F. Fung, X. Ma, H. F. Chau, *Practical issues in quantum-key-distribution postprocessing*, Phys. Rev. A **81**, 012318 (2010), arXiv:0910.0312.